STUDY COMPARISON · 5-DATASET DRAFT VS. 10-DATASET STUDY

Five versus ten: what actually changed

In August, this study ran on 5 UCI datasets. In September, 5 more were added, spanning signal processing, astronomy, and chemistry, and the full pipeline was rerun end to end. This is a number-by-number account of what moved, what held, and the one place the added scale overturned a published claim rather than just shifting it.

5-dataset (old) 10-dataset (new) finding reversed, not just shifted

Study scale

Doubling the datasets roughly doubled everything downstream of them, exactly as it should if nothing was broken by the extension.

Datasets	5	**10**
Raw per-seed runs	6,417	**12,837**
Seed-averaged method-condition rows	2,139	**4,279**
Failed runs (all `tabpfn_downsample` at r=0.50)	63	**123**
of which, Direction A	60	**120**
of which, clean condition	3	**3**

1 Overall robustness: CHR and mean Correction Gain

Most methods held their rank. `tabpfn_threshold` stayed the clear leader on both measures in both directions. The one real reshuffle: `tabpfn_downsample`'s Direction A harm rate dropped ten points (31.7% to 21.7%), while both DistPFN variants ticked up, so the two DistPFN variants, not downsample, are now unambiguously the worst on CHR in both directions.

	CHR, DIR. A		CHR, DIR. B		MEAN CG, DIR. A		MEAN CG, DI	
METHOD	OLD	NEW	OLD	NEW	OLD	NEW	OLD	
xgb_weighted	15.0%	**16.9%**	10.0%	**9.4%**	0.016	**0.020**	0.025	**0**
xgb_ros	18.8%	**18.8%**	12.5%	**12.5%**	0.013	**0.018**	0.024	**0**
xgb_smote	25.0%	**21.3%**	10.0%	**8.1%**	0.016	**0.024**	0.034	**0**
tabpfn_threshold	11.9%	**8.1%**	0.6%	**1.6%**	0.053	**0.061**	0.076	**0**
tabpfn_distpfn	23.1%	**23.4%**	28.8%	**31.9%**	0.005	**0.016**	0.011	**0**
tabpfn_distpfn_t	22.5%	**24.4%**	25.0%	**30.9%**	0.009	**0.021**	0.020	**0**
tabpfn_downsample	31.7%	**21.7%**	7.5%	**7.5%**	0.046	**0.051**	0.069	**0**

Two rows stayed exactly flat: `xgb_ros`'s CHR in both directions, and `tabpfn_downsample`'s Direction B CHR (7.5% in both the old and new run), the most stable numbers in the whole comparison.

2 Regret: how bad a failure is when it happens

This is where the CHR reshuffle above actually matters. In the old study, `tabpfn_downsample` had the higher Direction A failure rate of the pair. In the new study, that's reversed, `tabpfn_distpfn` now fails more often *and* more severely, in both directions.

	DIRECTION A		DIRECTION B	
METHOD	CHR	MEAN REGRET	CHR	MEAN REGRET
tabpfn_downsample · old	31.7%	0.0063	7.5%	0.0002
tabpfn_downsample · new	**21.7%**	**0.0045**	**7.5%**	**0.0003**
tabpfn_distpfn · old	23.1%	0.0089	28.8%	0.0104
tabpfn_distpfn · new	**23.4%**	**0.0073**	**31.9%**	**0.0172**

3 Direction asymmetry

Same shape, softer edges. Direction B still hurts the uncorrected baselines more than Direction A, that finding held exactly. The size of the gap narrowed slightly with more data.

	DIR. A DROP	DIR. B DROP	RATIO
xgb_none · old	0.030	0.083	2.8×
xgb_none · new	**0.032**	**0.079**	**2.5×**
tabpfn_none · old	0.022	0.055	2.5×
tabpfn_none · new	**0.037**	**0.067**	**1.8×**

4 Break-even share

Share of (dataset, r) combinations where a method never crossed zero through c=0.30. `tabpfn_threshold` gained the most ground here, up from tied-for-middle to tied for the Direction B lead alongside two XGBoost corrections.

	DIRECTION A		DIRECTION B	
METHOD	OLD	NEW	OLD	NEW
xgb_weighted	45%	**48%**	70%	**70%**
xgb_ros	55%	**55%**	70%	**70%**
xgb_smote	35%	**38%**	70%	**78%**
tabpfn_threshold	50%	**55%**	65%	**70%**
tabpfn_distpfn	30%	**38%**	30%	**35%**
tabpfn_distpfn_t	40%	**40%**	35%	**40%**
tabpfn_downsample	60%	**60%**	65%	**70%**

5 Prior Sensitivity Gap

The headline held: the oracle prior still does not reliably outperform the corrupted, observed one, negative for all three prior-dependent methods in both directions, in both studies. What reordered is *which* method has the largest gap. In the old study `tabpfn_threshold` had the smallest-magnitude PSG of the three. In the new study it has the largest.

	DIRECTION A		DIRECTION B	
METHOD	OLD	NEW	OLD	NEW
tabpfn_threshold	−0.008	**−0.013**	—	**−0.008**
tabpfn_distpfn	−0.014	**−0.010**	—	**−0.020**
tabpfn_distpfn_t	−0.015	**−0.010**	—	**−0.015**

At c=0.30 only, Direction A: `tabpfn_distpfn`'s gap widened from −0.060 (old) to −0.042 (new), still the widest of the three, but by a much narrower margin over the other two (−0.041, −0.039) than before. Direction B wasn't reported at this granularity in the old draft, marked — above.

6 Statistical significance

More data did what more data is supposed to do: every Friedman statistic grew, every p-value shrank by roughly a dozen orders of magnitude, and pairwise ambiguity mostly resolved itself.

	X^2 OLD	X^2 NEW	N OLD	N NEW
XGBoost · Dir. A	55.3	**111.1**	80	**160**
XGBoost · Dir. B	91.6	**198.9**	80	**160**
TabPFN · Dir. A	64.3	**173.2**	60	**120**
TabPFN · Dir. B	154.9	**332.7**	80	**160**

All four tests: $p < 10^{-11}$ (old) → ***$p < 10^{-23}$ (new)***. Non-significant pairwise comparisons, the pairs of methods statistics couldn't tell apart, dropped from **5** to **1**. The one that survived: `xgb_weighted` vs. `xgb_ros` under Direction B, unchanged from a genuine tie in both studies, not a rounding artifact.

THE ONE FINDING THAT DIDN'T JUST SHIFT, IT FLIPPED

Calibration

Old claim: under Direction A, DistPFN and DistPFN-T "join" tabpfn_downsample in crossing above the uncorrected baseline's Brier score and ECE once contamination reaches 20%, i.e. their calibration gets actively worse than doing nothing.

Rechecked cell by cell against the 10-dataset data: this is false. DistPFN and DistPFN-T never cross above baseline on either metric, in either direction, at any tested noise level. They are, instead, the two best-calibrated methods of any tested, correction or baseline, at nearly every noise level in both directions, only `tabpfn_downsample` ever overtakes them, and only at the single highest noise level under Direction B.

This means the same two methods hold both the worst Correction Harm Rate in the study (worst at getting the classification right) and the best calibration in the study (best at knowing how unsure to be about it). That's not a contradiction, discrimination and calibration are different failure modes, but it's a genuinely different, more interesting story than "these two corrections are just bad," which is what the 5-dataset draft said.

What held, and what moved

HELD EXACTLY

- `tabpfn_threshold` remains the most robust method on CHR and mean CG, both directions
- Same 5-vs-2 direction split: five methods fail more under Dir. A, the two DistPFN variants fail more under Dir. B
- PSG negative for all three prior-dependent methods, both directions, both studies
- `tabpfn_distpfn` has the highest regret in both directions in both studies
- Direction B degrades the uncorrected baselines more than Direction A, in both studies

SHIFTED, SAME DIRECTION

- Most CHR / mean CG values moved a few points, no rank changes except downsample vs. the DistPFN pair
- Every p-value shrank by roughly 10^{12}, more power, not a different result
- Non-significant pairs: 5 → 1
- PSG magnitudes reordered among the three methods, sign and overall conclusion unchanged
- Direction-asymmetry ratio softened (2.8× → 2.5× for XGBoost)

Every figure above traces to a specific commit in this project's git history, comparing the 5-dataset paper draft against the current 10-dataset sections and results/*.csv. Compiled 2 September 2026.